General Theory of Layered Authority

Mathematical Spine for the Six-Layer Fixed Authority Cosmology

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Abstract

This paper formalizes the six-layer fixed authority cosmology as a typed finite-authority stack. The central claim is not that all truth is finite, nor that every domain already has a known finite verifier. The central claim is narrower:

$$\boxed{\text{Operational authority exists only as terminalized, certified finite descent.}}$$

The governing normal form is the Kleisli composite

$$\boxed{\Pi^\sharp = \tau^\sharp \circ_K \mathrm{FOC}^\sharp \circ_K D^\sharp \circ_K \rho^\sharp \circ_K \mathcal{R}^\sharp : X \to \mathbb{T} + \mathbb{F}.}$$

A proposal is authority-bearing exactly when this composite returns an authority terminal:

$$\boxed{\mathsf{Auth}(x) \iff \exists t \in \mathbb{T}_{\mathsf{auth}}, \Pi^\sharp(x) = \mathsf{inl}(t).}$$

The six layers are not six sources of authority. They are six conserved obligations: finite obstruction, finite descent authority, certified representation transport, completion comparison, continuum shadow, and residual closure.

1 Governing Rule

The stack begins with the primitive rule:

$$\boxed{\text{Difference is primitive; sameness is certified zero-difference.}}$$

Thus identity, equivalence, transport, completion, and closure are never silently assumed. They must be certified inside a declared regime. Failure at any layer is not automatically disproof. It is one of:

- a typed refusal,
- an unresolved boundary,
- a non-authority terminal,
- a demand for fresh descent,
- or a licensed residual.

The operational slogan is:

$$\boxed{\text{No descent, no authority. No certificate, no transport. No terminal, no promotion.}}$$

2 The Base Arena: FAS

Let **FAS** denote the 2-category of finite authority systems.

2.1 Objects

Objects are finite authority pipelines

$$\mathcal{F} = (X, \mathcal{R}, \rho, D, V, \tau, \mathbb{T}, \mathbb{F}, \mathsf{Cert}, \mathsf{Ref}).$$

Here:

Symbol	Role
X	proposal/input domain
$\mathcal{R}$	representation gate
ρ	routing map
D	finite descent compiler
V	verifier
FOC	obstruction-calculus realization of V, when available
τ	terminalizer
$\mathbb{T}$	closed terminal outcomes
$\mathbb{F}$	typed refusals
Cert	admissible certificates
Ref	refusal semantics

2.2 1-cells

A 1-cell

$$F : \mathcal{F} \to \mathcal{G}$$

is a lawful pipeline or representation morphism. If it claims to preserve authority, it must carry a descent-compatibility 2-cell:

$$\boxed{\chi_F : D_{\mathcal{G}} F \Rightarrow \bar{F} D_{\mathcal{F}}.}$$

No anonymous representation shift is authority-preserving. A lossy morphism must return a loss ledger:

$$\mathsf{Loss}(F) = (\mathsf{erased}, \mathsf{identified}, \mathsf{approximated}, \mathsf{unresolved}).$$

2.3 2-cells

A 2-cell is a certificate witness

$$\alpha : F \Rightarrow G$$

that preserves the relevant representation, descent, verification, and terminal semantics up to declared refinement or typed refusal.

The fundamental transport law is:

$$\boxed{\text{No pasted certificate, no authority transport.}}$$

3 The Exception Monad and Kleisli Pipeline

Authority stages are partial in ordinary language, but total in the exception/refusal monad

$$M(Y) = Y + \mathbb{F}.$$

Each stage is a Kleisli arrow:

$$\mathcal{R}^\sharp : X \to \mathcal{C} + \mathbb{F},$$

$$\rho^\sharp : \mathcal{C} \to \mathcal{S} + \mathbb{F},$$

$$D^\sharp : \mathcal{S} \to \mathcal{P} + \mathbb{F},$$

$$V^\sharp : \mathcal{P} \to \mathbb{V} + \mathbb{F},$$

$$\tau^\sharp : \mathbb{V} \to \mathbb{T} + \mathbb{F}.$$

The total terminal-preserving procedure is

$$\boxed{\Pi_S^\sharp = \tau^\sharp \circ_K V^\sharp \circ_K D^\sharp \circ_K \rho^\sharp \circ_K \mathcal{R}^\sharp.}$$

When V is realized by finite obstruction calculus, the strongest form is

$$\boxed{\Pi_S^\sharp = \tau^\sharp \circ_K \mathrm{FOC}^\sharp \circ_K D^\sharp \circ_K \rho^\sharp \circ_K \mathcal{R}^\sharp.}$$

4 Soundness Class

A system S belongs to $\mathbf{AuthSys}_{\text{sound}}$ iff it satisfies the anti-cheat axioms below.

Axiom 4.1 (Finite operational signature)**.** *S has a finite input signature $\mathcal{P}$, an admissible representation grammar $\mathcal{G}$, and a finitely enumerable set of admissible representation morphisms $\mathcal{M}$.*

Axiom 4.2 (Representation necessity)**.**

$$\boxed{\mathsf{Auth}_S(x) \Rightarrow \mathcal{R}(x) \downarrow .}$$

Axiom 4.3 (Certified representation transport)**.** *Every authority-preserving representation change requires a typed morphism and a certificate χ_F.*

Axiom 4.4 (Finite replay requirement)**.**

$$\boxed{\mathsf{Promote}_S(x) \Rightarrow D(\rho(\mathcal{R}(x))) \downarrow \wedge V(D(\rho(\mathcal{R}(x)))) \downarrow .}$$

Axiom 4.5 (Weak terminal totality)**.** *Every completed verification attempt terminates into the closed terminal set $\mathbb{T}$.*

Axiom 4.6 (Non-promotion of non-authority states)**.**

$$\boxed{\tau(V(D(\rho(\mathcal{R}(x))))) \notin \mathbb{T}_{\mathsf{auth}} \Rightarrow \neg\mathsf{Auth}_S(x).}$$

Axiom 4.7 (Authority requires promotion path)**.**

$$\boxed{\mathsf{Auth}_S(x) \Rightarrow \mathsf{Promote}_S(x).}$$

Axiom 4.8 (No shadow veto)**.**

$$\boxed{\exists t \in \mathbb{T}_{\mathsf{auth}}, \Pi_S^\sharp(x) = \mathsf{inl}(t) \Rightarrow \mathsf{Auth}_S(x).}$$

This axiom forbids a hidden post-terminal veto layer.

5 Authority Characterization Theorem

Theorem 5.1. *If* $S \in \mathbf{AuthSys}_{\text{sound}}$*, then*

$$\boxed{\forall x, \quad \mathsf{Auth}_S(x) \iff \exists t \in \mathbb{T}_{\mathsf{auth}}, \Pi_S^\sharp(x) = \mathsf{inl}(t).}$$

Proof sketch. Necessity follows from A1, A3, A5, and A6: authority requires representation, finite descent, replayable verification, and an authority-bearing terminal. Sufficiency follows from A7: if the declared pipeline returns an authority terminal, the system must recognize it as authority.

Thus authority is not a property of a proposal, representation, completion, or continuum object by itself. Authority is the terminal status of the declared composite.

Full proof: see Proof Archive, Sections 11–18. □

6 Homological Verifier Spine: Finite Obstruction Calculus

The local verifier is a finite obstruction calculus.

An obstruction packet is

$$\mathcal{M} = (\mathcal{T}, \mathcal{C}, \rho, C^\bullet, \omega, p, \mathcal{K}, \Lambda)$$

with finite cochain complex

$$\boxed{C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2, \qquad d_1 d_0 = 0.}$$

Define:

$$B^1 = \operatorname{im}(d_0)$$

$$Q^1 = C^1 / B^1$$

$$\bar{d}_1 : Q^1 \to C^2, \qquad \bar{d}_1([x]) = d_1 x$$

$$H^1 = \ker(\bar{d}_1) \cong \ker(d_1) / \operatorname{im}(d_0).$$

The obstruction norm is

$$\boxed{\Phi([x]) = \inf_{\alpha \in C^0} \|x - d_0 \alpha\|_{1,\omega}.}$$

Interpretation:

Object	Meaning
B^1	exact repair image
Q^1	quotient obstruction space
Φ	obstruction mass
$\bar{d}_1$	closure defect
H^1	closed non-exact residual sector
$w = (x, y, c)$	context-visible non-congruence witness
$\mathcal{K}$	replay certificate class
Λ	lift/refusal/budget discipline

Theorem 6.1 (Root Authority–Obstruction Factorization)**.** *For a FOC-refined sound system and a descended packet $p(x) \in C^1$:*

$$\boxed{[p(x)] = 0 \in Q^1 \Rightarrow \mathsf{ExactAuthority}(x).}$$

$$\boxed{[p(x)] \neq 0 \in Q^1 \Rightarrow \neg\mathsf{ExactAuthority}(x).}$$

Nonzero obstruction may still terminalize as survivor, licensed lift, hold, budget, open obstruction, or typed refusal. It does not authorize exactness.

Proof. Full proof: Proof Archive, Section 11. □

7 FDA–FOC Refinement

Finite Descent Authority supplies the global authority doctrine:

$$\boxed{\mathsf{Auth} = \tau \circ V \circ D.}$$

FOC supplies the local verifier. The refinement is lawful only when the FDA packet is FOC-realized.

A descent packet is the dependent record

$$P_s = (K_s, E_s, V_s, \tau_s, \Lambda_s).$$

FOC realizes V_s when it supplies an obstruction packet

$$\mathcal{G}_s = (K_s, \mathcal{C}_s, \rho_s, C_s^\bullet, E_s, \mathcal{O}_s, \mathcal{K}_s, \Lambda_s)$$

with:

1. same carrier,
2. same candidate class,
3. replayable certificate adequacy,
4. extensional verifier agreement,
5. output-fiber preservation,
6. residual typing,

7. terminal compatibility.

Theorem 7.1 (FDA–FOC Refinement)**.** *If every descended packet used by S is FOC-realized, then*

$$\boxed{\tau_s^\sharp \circ_K V_s^\sharp = \tau_s^\sharp \circ_K \mathrm{FOC}_s^\sharp}$$

on all FOC-realized packets. Consequently,

$$\boxed{\mathsf{Auth}_S(x) \iff \exists t \in \mathbb{T}_{\mathsf{auth}}, (\tau^\sharp \circ_K \mathrm{FOC}^\sharp \circ_K D^\sharp \circ_K \rho^\sharp \circ_K \mathcal{R}^\sharp)(x) = \mathsf{inl}(t).}$$

Proof. Full proof: Proof Archive, Section 12. □

8 The Six-Layer Doctrine Tower

The full tower is

$$\boxed{\mathsf{GTOC} \xrightarrow{i_0} \mathsf{GTFDA} \xrightarrow{i_1} \mathsf{GTMUR} \xrightarrow{i_2} \mathsf{GTOR} \xrightarrow{i_3} \mathsf{GTTC} \xrightarrow{i_4} \mathsf{GTTA}.}$$

Under the minimal no-identification doctrine,

$$\boxed{A \equiv B \text{ only if } A \text{ and } B \text{ have the same type, inputs, outputs, failure terminals, and laws.}}$$

Therefore:

$$\boxed{N_{\mathrm{meta}} = 6.}$$

8.1 The six conserved roles

- **GTOC / Obstruction:** objects $Q^1, \Phi, \Gamma, H^1, w, \mathcal{K}, \Lambda$; failure terminal `NoObstructionKernel`.
- **GTFDA / Authority:** terminalized finite descent $\Pi^\sharp : X \to \mathbb{T} + \mathbb{F}$; failure terminal `NoAuthoritySemantics`.
- **GTMUR / Transport:** descent-compatible certificate $\chi_F : D_B F \Rightarrow \bar{F} D_A$; failure terminal `TransportViolation`.
- **GTOR / Completion comparison:** comparison objects, limits, pullbacks, and weak-terminal envelopes; failure terminal `NoCompletionComparison`.
- **GTTC / Continuum shadow:** finite shadow tower $\{A_\lambda, \pi_\lambda, \chi_{\lambda\mu}\}$; failure terminal `NoFiniteShadow`.
- **GTTA / Residual closure:** closure residue terminal $\mathrm{Close}_\Lambda(X)$ and $\mathsf{TypedAnomalon}_\Lambda(X)$; failure terminal `NoResidualTerminal`.

None of these roles may be deleted without losing a type-specific obligation.

9 Higher-Categorical Normal Form

There is a bicategory of doctrine stacks

$$\mathbf{DocStack}_{\partial}$$

whose objects are doctrine layers or stacks

$$\mathcal{D} = (\mathcal{C}, \mathcal{R}, \rho, D, V, \tau, \mathbb{T}, \mathbb{F}, \chi, \Lambda).$$

A lawful 1-morphism

$$F : \mathcal{D} \to \mathcal{E}$$

may embed, refine, quotient, approximate, lift, or reinterpret. If it transports authority, it must exhibit

$$\boxed{\chi_F : D_{\mathcal{E}} F \Rightarrow \bar{F} D_{\mathcal{D}}.}$$

A 2-morphism

$$\alpha : F \Rightarrow G$$

is a certified comparison preserving terminalized descent. The core coherence condition is

$$\boxed{\tau_{\mathcal{E}} \, \mathrm{FOC}_{\mathcal{E}} \, D_{\mathcal{E}} F \Rightarrow \tau_{\mathcal{E}} \bar{F} \, \mathrm{FOC}_{\mathcal{D}} \, D_{\mathcal{D}}.}$$

Theorem 9.1 (Higher-Categorical Normal Form)**.** *The six-layer tower admits the normal form*

$$\boxed{\Pi^{\sharp} = \tau^{\sharp} \circ_K \mathrm{FOC}^{\sharp} \circ_K D^{\sharp} \circ_K \rho^{\sharp} \circ_K \mathcal{R}^{\sharp}.}$$

The upper layers do not alter this authority normal form. They add transport, comparison, continuum shadow, and residual closure obligations around it.

Proof. Full proof: Proof Archive, Section 30. □

10 Dependent Type and HoTT Reading

The stack can be read as a proof-relevant dependent type theory for authority.

10.1 Authority as a dependent type

$$\boxed{\mathsf{Authority}(x) := \Sigma(t : \mathbb{T}_{\mathsf{auth}})\big(\Pi^{\sharp}(x) = \mathsf{inl}(t)\big).}$$

An authority result is not merely a Boolean. It is a dependent pair (t, π), where t is an authority terminal and π proves that the pipeline produced it.

10.2 Refusal as a dependent type

$$\boxed{\mathsf{Refuse}(x) := \Sigma(\phi : \mathbb{F})\big(\Pi^{\sharp}(x) = \mathsf{inr}(\phi)\big).}$$

Thus non-authority is not automatically contradiction. It is often an inhabited refusal type.

10.3 Descent packet as dependent record

$$\boxed{\begin{aligned}\mathsf{Packet}(s) := {} & \Sigma(K : \mathsf{Carrier}(s))\ \Sigma(E : \mathsf{EvalClass}(K)) \\ & \Sigma(V : \mathsf{Verifier}(K, E))\ \Sigma(\tau : \mathsf{Terminalizer}(V)) \\ & \mathsf{LiftDiscipline}(K, E, V, \tau).\end{aligned}}$$

Deleting a field makes the packet ill-typed.

10.4 HoTT relation

HoTT transports evidence along paths. FDA transports authority only along certified morphisms:

$$\boxed{\mathsf{TransportAuth}_F(x) \Rightarrow \chi_F(x) \downarrow .}$$

This is a stricter, finite, replayable analogue of transport. Univalence is not accepted as unrestricted authority transport. The admissible analogue is certified χ-univalence: equivalence preserves authority only when equipped with a descent-compatible certificate.

Full discussion: Proof Archive, Sections 29 and 32.

11 Upper-Layer Non-Creation

No layer above GTFDA creates authority.

$$\boxed{L > \mathsf{GTFDA} \Rightarrow\ L \text{ may organize, transport, compare, complete, shadow, or expose residue, but may not create authority.}}$$

Equivalently:

$$\boxed{X \in L \not\Rightarrow \mathsf{Auth}(X).}$$

Authority may appear in an upper layer only when there is certified descent back to the finite authority normal form.

11.1 GTMUR

Representation does not authorize:

$$\boxed{\mathcal{R}(x) \not\Rightarrow \mathsf{Auth}(x).}$$

Transport requires χ_F. Loss requires a loss ledger.

11.2 GTOR

Completion does not authorize:

$$\boxed{X \in \mathrm{Comp}(\mathsf{GTMUR}) \not\Rightarrow \mathsf{Auth}(X).}$$

Products, pullbacks, limits, comparison objects, and weak terminal envelopes require descent certificates before they become authority-bearing.

11.3 GTTC

Continuum coherence does not authorize:

$$\boxed{X \simeq \lim_{\lambda \in \Lambda} i_3(A_\lambda) \not\Rightarrow \mathsf{Auth}_{\mathsf{GTTC}}(X).}$$

Authority in GTTC requires a cofinal system of authorized finite shadows:

$$\boxed{\exists \Lambda_0 \subseteq \Lambda \text{ cofinal such that } \forall \lambda \in \Lambda_0, \mathsf{Auth}_{\mathsf{GTOR}}(A_\lambda) \downarrow \text{ and } \chi_{\lambda\mu} \downarrow .}$$

11.4 GTTA

The anomalon is typed non-absorption, not origin:

$$\boxed{\mathsf{TypedAnomalon}_\Lambda(X) \not\Rightarrow \mathsf{OriginOfDifference}(X).}$$

A typed anomalon exists only after a licensed closure attempt:

$$\boxed{\mathrm{Close}_\Lambda(X) = \mathsf{NonAbsorb}(a) \Rightarrow \mathsf{TypedAnomalon}_\Lambda(X) = a.}$$

Full proofs: Proof Archive, Sections 20, 24, 28, and 31.

12 The Three Counts of N

The framework has three distinct finite counts.

12.1 Core pipeline count

$$\boxed{N_{\mathrm{FDA}} = 5.}$$

This counts the operational authority stages:

$$(\mathcal{R}, \rho, D, V, \tau).$$

12.2 Meta-role count

$$\boxed{N_{\mathrm{meta}} = 6.}$$

This counts the irreducible tower obligations:

$$(\mathsf{O}, \mathsf{A}, \mathsf{T}, \mathsf{C}, \mathsf{S}, \mathsf{R}).$$

12.3 Proof-certificate length

$$\boxed{N_{\mathrm{proof}}(\pi) = |\pi|.}$$

This counts a particular finite proof certificate, if such a certificate exists.
The key separation is:

$$\boxed{N_{\mathrm{meta}} \text{ counts the machine.}}$$

$$\boxed{N_{\text{proof}} \text{ counts the certificate the machine verifies.}}$$

Full proof: Proof Archive, Section 27.

13 Main Theorem: Authority Conservation

Theorem 13.1 (Authority Conservation)**.** *Let*

$$\mathcal{S}_6 = (\mathsf{GTOC}, \mathsf{GTFDA}, \mathsf{GTMUR}, \mathsf{GTOR}, \mathsf{GTTC}, \mathsf{GTTA})$$

be a sound six-layer doctrine stack in $\mathbf{DocStack}_\partial$*. Then, for every proposal* x*:*

$$\boxed{\mathsf{Auth}(x) \iff \exists t \in \mathbb{T}_{\mathsf{auth}}, (\tau^\sharp \circ_K \mathrm{FOC}^\sharp \circ_K D^\sharp \circ_K \rho^\sharp \circ_K \mathcal{R}^\sharp)(x) = \mathsf{inl}(t).}$$

Moreover, any operation in GTMUR*,* GTOR*,* GTTC*, or* GTTA *preserves authority only by exhibiting a descent-compatible certificate back into this normal form.*

Proof sketch.

1. A1–A7 give the authority characterization theorem.
2. The descent compiler supplies well-typed packets $P_s = (K_s, E_s, V_s, \tau_s, \Lambda_s)$.
3. FOC realizes V on obstruction-realized packets.
4. Q^1 separates exact authority from obstruction at the verification stage.
5. χ_F is required for representation transport.
6. Completion, continuum, and closure layers are authority-noncreating by the upper-layer non-creation theorem.
7. Therefore all legitimate authority in the six-layer stack factors through the same finite terminal normal form.

Full proof dependencies: Proof Archive, Sections 11–34. □

14 Publication Split

The strongest publication order is two papers plus a proof archive.

14.1 Paper I — Layer Factorization and Authority Conservation

This paper should contain:

1. **FAS** and $\mathbf{DocStack}_\partial$,
2. the exception/Kleisli normal form,
3. A0–A7,
4. Authority Characterization,
5. the six meta-roles,

6. the upper-layer non-creation theorem,
7. Authority Conservation.

Its core formula is:

$$\boxed{\mathsf{Auth} = \tau \circ \mathrm{FOC} \circ D \circ \rho \circ \mathcal{R}.}$$

14.2 Paper II — Finite Obstruction and Descent Authority

This paper should contain:

1. finite cochain packets,
2. $Q^1 = C^1 / \operatorname{im}(d_0)$,
3. $H^1 = \ker(\bar{d}_1)$,
4. Φ and certificates,
5. FOC realization of FDA verifier packets,
6. exact authority as zero obstruction,
7. nonzero obstruction as survivor/open/lift/refusal terminal.

Its core formula is:

$$\boxed{[p] = 0 \in Q^1 \iff \mathsf{ExactAuthority}(p)}$$

on descended, FOC-realized packets.

14.3 Companion — Proof Archive

The proof archive should hold all long lemmas, applications, caveats, and future proof targets. That keeps the main paper readable.

15 Non-Claims

This spine does not claim:

1. that every truth is finite;
2. that every domain already has a known descent map;
3. that global geometry, completion, or continuum structure is useless;
4. that the anomalon is the origin of difference;
5. that representation, completion, or coherence creates authority;
6. that unsolved mathematical problems have already been solved;
7. that a proof certificate exists before it is produced.

It claims only:

Operational authority, as defined here, requires representation, routing, finite descent, verification, terminalization, and certified transport where applicable.

16 Closing Normal Form

The complete spine fits on one line:

FAS supplies the arena;
FOC supplies the finite homological verifier;
$\mathbf{DocStack}_\partial$ supplies the higher-categorical tower;
$\mathsf{Authority}(x) = \Sigma(t : \mathbb{T}_{\mathsf{auth}})\big(\Pi^\sharp(x) = \mathsf{inl}(t)\big)$
supplies the HoTT/dependent-type reading;
$\mathsf{GTMUR}, \mathsf{GTOR}, \mathsf{GTTC}, \mathsf{GTTA}$ organize without creating authority.

There may be many worlds, representations, completions, and residues. But authority must descend.